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EcoLogits: Evaluate the Environmental Impacts of Generative AI

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Summary

Recent advancements in generative AI (GenAI), particularly Large Language Models (LLMs), have led to significant increases in computational power and energy demands. These models are not only becoming larger and more complex but are also being widely deployed across various platforms, reaching millions of users and developers. This widespread use has escalated the overall energy consumption and environmental impact associated with these technologies.

To address these concerns, we introduce EcoLogits, an open-source Python package designed to evaluate the environmental footprint of using GenAI models. EcoLogits estimates energy consumption and multi-criteria environmental impacts of GenAI API requests using a bottom-up Life Cycle Assessment methodology, considering both direct energy use and hardware production impacts. EcoLogits adheres to high standards of open-source practices, including versioning, automated testing, and comprehensive code documentation.

Statement of need

While the recent developments in GenAI offer tools of unprecedented quality, there is a general trend of increased energy consumption associated with these technologies (Vries, 2023). It is a well-known fact that training large neural networks yields non-negligible carbon emissions (Lacoste et al., 2019; Patterson et al., 2021). Some industry actors communicate about the emissions of training their own models, such as Meta that reported the carbon emissions of Llama models (AI@Meta, 2024; Touvron, Lavril, et al., 2023; Touvron, Martin, et al., 2023).

The major difference in recent years is that GenAI models are not only large models but are also widely used by millions of people through web application such as OpenAI's ChatGPT or via APIs. The literature agrees on the fact that the share of carbon emissions from the inference is increasing (Desislavov et al., 2023; Patterson et al., 2021) although these proportions vary among providers, with AWS reporting 80-90%, Google 60% and Meta 33% (Luccioni et al., 2024).

Moreover, while energy consumption is important, it is alone insufficient for a comprehensive evaluation of the environmental impacts of training and inference. It is crucial to include embodied impacts that come from hardware production. For LLMs such as BLOOM 176B, accounting for the embodied impacts increased the total carbon emissions by 22% for the training phase of the model (Luccioni et al., 2022). Environmental impacts evaluations of AI models should be more transparent and follow Life Cycle Assessment methodologies to be complete (Ligozat et al., 2022).

In parallel to the large deployment of energy-consuming models in the industry, there is a growing need for tools that automatically approximate their environmental impacts. While tools like CodeCarbon or Zeus (Courty et al., 2024; You et al., 2023) can track energy consumption and carbon emissions for locally-deployed models, estimating impacts becomes necessary when

using external APIs from GenAI service providers. EcoLogits is the first Python library to approximate the environmental impacts of GenAI models inference via Python client calls. By adding `EcoLogits.init()` to their codebase, users can access an estimation of the energy consumption and the environmental impacts of their requests.

Methodology

The **Attributional Life Cycle Assessment (A-LCA)** methodology defined by the ISO 14044 standard (Boustead, 1996; Hunt et al., 1996; Klöpffer, 1997) is used to properly estimate potential environmental impacts of products, processes, projects or services. In EcoLogits, we apply this methodology to estimate the impacts of a request to an GenAI inference service. The methodology encompasses multi-phases and multi-criteria environmental impacts.

In our approach we consider the usage phase accounting for the direct energy consumption (Ilyas Moutawwakil, 2023a, 2023b) and the embodied phase accounting for the hardware production, including raw material extraction, manufacturing and transportation (Simon et al., 2024). We do not include the end-of-life phase for lack of data and transparency on e-waste collection and recycling (Bordage et al., 2021; Forti et al., 2020). EcoLogits reports an estimation of the direct energy consumption in addition to other environmental impacts criteria such as *Global Warming Potential* (Stocker et al., 2013), *Abiotic Depletion Potential* (Oers et al., 2020) and *Primary Energy* consumption (Frischknecht et al., 2015). Assessing multiple impact criteria helps prevent pollution shifting that can result from optimizing for a single criterion.

To estimate the impacts of a GenAI service we use a **bottom-up modeling approach**, i.e. we assess the environmental impacts of all components that compose the service individually and then aggregate and allocate these impacts to a request. The scope of our methodology encompasses cloud compute primitives and data center extra equipment such as cooling (see Figure 1). We focus our study on high-performance GPU-accelerated cloud instances since they are commonly used for GenAI inference tasks. We do not account for other impacts related to model training, networking transfers or end-user devices.

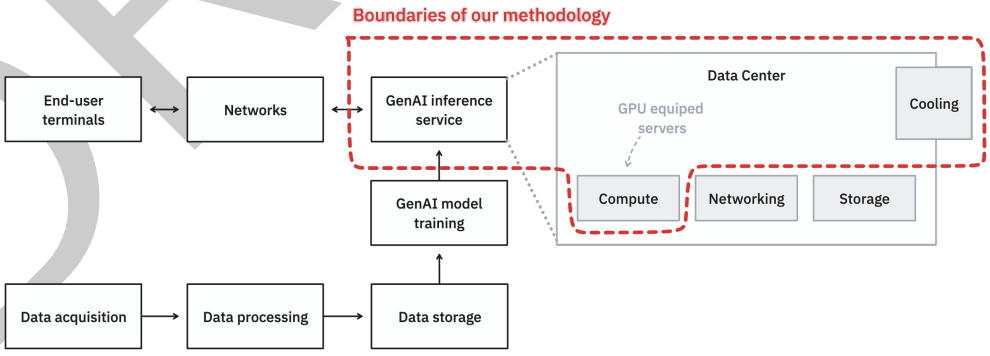

Figure 1: Scope and boundaries of our methodology focused on compute primitives

Discussions

Our approach for estimating the environmental impacts of GenAI inference aligns with industry standards and focuses on real-world deployments. As the field is constantly evolving, we are committed to refining our methodology to keep up with new model architectures, optimization techniques, and hardware innovations. Our tool enables GenAI practitioners to evaluate and

compare these models based on their environmental footprints. In the future, we plan to expand our scope to include more stages of the model life cycle and new modalities.

One challenge we face is dealing with proprietary models from companies that do not share details about their models and infrastructure. In these cases, we have to make educated guesses, such as estimating the number of parameters in models and which hardware is being used in production. We document these assumptions carefully and update them regularly as new models are released.

Finally, we want to mention deeper concerns about impact assessment of GenAI. Our approach, while robust, does not capture the full range of environmental impacts associated with GenAI technologies. To get a complete picture, indirect impacts and potential rebound effects must be considered. Pursuing 'Green AI' means more than just making models smaller and more efficient - it requires a holistic view of the environmental changes driven by this technology, as well as its broader societal and service-level implications.

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